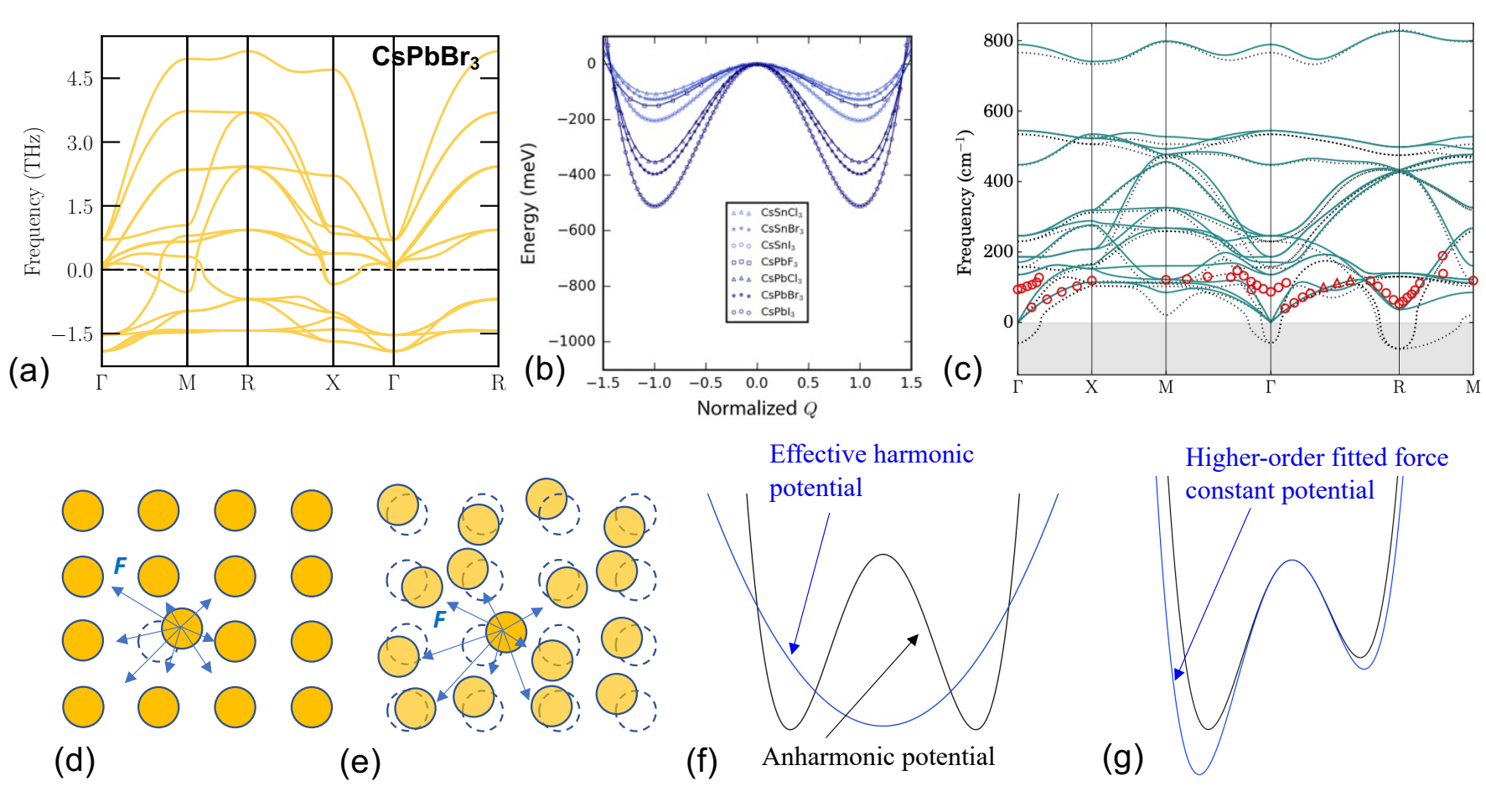

CsPbBr3
Frequency (THz)
4.5
3.0
1.5
0.0
−1.5
Γ
M
R
X
Γ
R
(a)
Energy (meV)
0
−200
−400
−600
−800
−1000
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Normalized Q
CsSnCl3
CsSnBr3
CsSnI3
CsPbF3
CsPbCl3
CsPbBr3
CsPbI3
(b)
Frequency (cm−1)
800
600
400
200
0
Γ
X
M
Γ
R
M
(c)
F
(d)
F
(e)
Effective harmonic potential
Anharmonic potential
(f)
Higher-order fitted force constant potential
(g)